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The onion universe as a viable cosmological model, a geometrical analysis

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Abstract

The onion universe is proposed as an alternative cosmological model offering a geometrical approach to some of the theoretical challenges faced by ΛCDM. It assumes a closed spatial geometry and describes the evolution of the Universe through a scale factor derived solely from geometric arguments within the FLRW metric. Predictions for cosmological distance measures in this model are tested against Type Ia supernova and baryon acoustic oscillation data, allowing constraints to be placed on three cosmological parameters: H_0, r_d, $\Omega_{\Lambda,0}$. The parameter estimation suggests a preference for a hyperboloidal spacetime geometry within the onion model. A χ^2 test is performed for both the onion and ΛCDM models, showing equivalent fits to observational data. The geometrical framework adopted in this work, provides no statistically significant distinction between the models.

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1 Introduction

In the present day, a great consensus exists along the physics community that we live in a flat universe with accelerated expansion, described by a spatial Euclidean geometry. The accelerated expansion rate would be explained by *dark energy* (DE), a mysterious fluid with negative pressure consistent with Einstein's *cosmological constant* Λ, dominating the dynamics of the universe. In this context, this text aims to present an alternative, the **onion model** (§§1.3). This work encompasses a study of the viability of its predictions regarding the geometry of spacetime, by comparing them with the most common cosmological observables.

1.1 Cosmology today

Just a year after the publishing of his *General Relativity*, Einstein began laying the foundations of the cosmological model [1]. He stated that the Universe is homogeneous at large scales, and proposed a matter-filled space, having to include the cosmological constant in his General Relativity (GR) field equations to provide that it was static, a wrong assumption of his model. In the 1920's, Friedmann and Lemaitre [2], [3], [4] introduced a varied collection of GR-based models until 1929 when Hubble [5] discovered direct evidence for the expansion of the Universe. Throughout the 1930's to the 1980's a succession of discoveries (such as the Cosmic Microwave Background (CMB) [6]) were followed with the proposal of diverse theories that comprise the currently accepted cosmological model. These include the establishment of inflationary theories, cold dark matter (CDM) models and the prediction of a spatially flat universe. In the 1990's, two independent groups conducted research on Type Ia supernovae (SNe) [7], [8], achieving distance estimates for higher-redshift objects than ever before. Their results pointed towards models with an accelerated expansion of the Universe.

Thus, we arrived to ΛCDM as the prevailing cosmological model. This framework consists of a flat universe (with space curvature k=0 (1)), whose components can be broken down into: baryonic matter and radiation (which nowadays constitute about 5% of the total energy density); dark matter (about 25%); and mainly dark energy ($\sim$70%), associated with the cosmological constant [9]. This components evolve according to the laws of GR and are described by the Friedmann equations, as we will discuss in §2. The foundations of this theory are constructed on the *Cosmological Principle*, that states the isotropy and homogeneity of space at sufficiently large scales (§§2.1). Under this premise, the metric that describes the geometry of spacetime (ST) is the *Friedmann-Lemaître-Robertson-Walker* (FLRW) metric, given by:

$$ds^2 = c^2dt^2 - R(t)^2\left[\frac{dr^2}{1-kr^2} + r^2(d\theta^2 + \sin^2\theta d\varphi^2)\right] \tag{1}$$

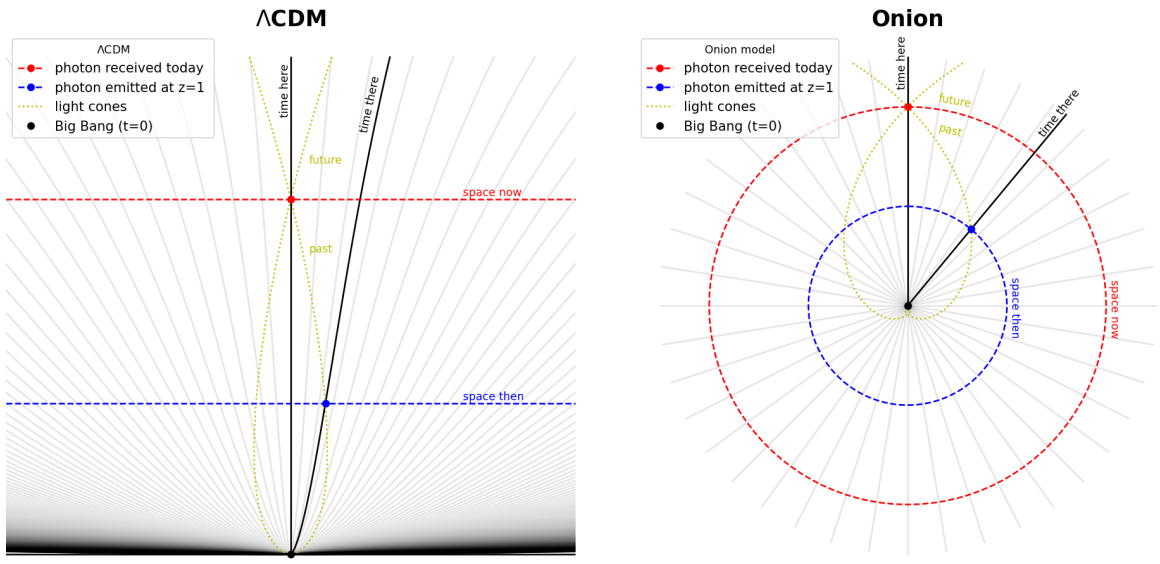
where *R(t)* is the scale factor as defined in (2). *k* denotes the curvature of space, and can take values; k>0 for a closed universe, k=0 for a flat universe, and k<0 for an open universe.

Nevertheless, several theoretical and observational challenges arise inside ΛCDM [10], [11], which motivate the search for an alternative. Some of this include: the *horizon problem*, which appears from the observed homogeneity of the Universe despite distant regions not being casually connected after the Big Bang (BB) (nevertheless, this problem is tackled by inflationary theories [12]); or the *flatness problem*, that arises from the apparent flatness of the Universe at present time, which requires a fine-tuning of the conditions of the early Universe; amongst others. We also want to draw attention to some of the astrophysical signals that could point towards the need of a revision of ΛCDM [10] (although they are not specifically addressed in this work): the *Hubble tension* (a 4σ incompatibility between the values for the Hubble constant measured locally and from the CMB data), the *CMB anisotropy anomalies* (the power spectrum of the CMB anisotropies alone favours a closed universe over a flat one [9]), the *Lithium problem* (less Lithium is observed than predicted by Big Bang Nucleosynthesis (BBN)), the *growth tension* (incompatibilities between observed structure at high redshift and predictions made by ΛCDM models), etc...

1.2 Physical motivation

In an effort to provide an alternative framework to ΛCDM, one that sets a new starting point and seeks to circumvent these issues, in this work we will explore the **onion universe** model. This paradigm is also based on the standard *Friedmann-Lemaître-Robertson-Walker* metric that describes any homogeneous and isotropic space [13], [14], but it postulates a *closed* geometry for space (k=1). The motivation for this proposal lies within a *generalization of the Cosmological Principle*; isotropy around the BB requires a closed geometry, as argued below (§§1.3). In a spatially flat geometry, such as ΛCDM, time is a semi-infinite coordinate that flows perpendicular to the Euclidean slices that represent different time stamps (fig. 1), being evidently anisotropic even around the BB (time exclusively flows *upwards* from the BB). Conversely, in a spatially closed universe, time appears as a radial coordinate that emerges isotropically from the BB. The Universe in this model is formed by a succession of concentric 3-hyperspheres of finite volume that correspond to consecutive instants in time, much like the layers of an onion, which grow following a scale factor $R(t)$ (3). This model can be understood as a generalization of *coasting cosmologies*, a family of alternative cosmologies well studied in the literature (with scale factor $R(t) \propto ct$) and first proposed by *Kolb, 1989* [15].

The main conceptual implication of a geometry of such type is a generalization of the *Cosmological Principle*. Such a proposal entails the complete homogeneity of spacetime, in the sense that any point of ST could host a BB around which the spacial slices would grow. This stands in contrasts to flat geometries, where each space slice is infinite and time extends perpendicularly to them, implying that half of spacetime corresponds to t<0 and therefore is not physical. Furthermore, as previously mentioned, a closed universe is isotropic around the BB; therefore extending both homogeneity and isotropy beyond space, to include spacetime.



(a) ΛCDM. Infinite flat space slices.

(b) Onion model. Finite closed space slices.

Figure 1: 1+1D spacetime diagrams. The observation (*red dot*) of a distant object at redshift z = 1 (*blue dot*) is used to illustrate the main properties of the standard ΛCDM cosmology (left) and a closed onion universe (right). The world lines of observers comoving with the expansion (*grey lines*) start at a particular event in spacetime, the BB (*black dot*), and propagate towards the future; the time coordinate measured by our clock (here) and that of the distant object (there) are shown as black solid lines. On the other hand, the spatial extent of the Universe at the present day (z = 0, red) and the time when the photons were emitted (z = 1, blue) are plotted as dashed lines. Space is flat in ΛCDM, and time flows upwards in the vertical direction (space slices are piled upwards). In our model, space is curved (concentric hyperspheres enclosing the BB), and time is a radial coordinate.

1.3 Onion model

The relation between the *cosmic time t* and the evolution of space slices in a FLRW metric is given by the **scale factor R(t)**, defined by:

$$\vec{X} = R(t) \cdot \vec{r} \tag{2}$$

where $\vec{X}$ corresponds to the physical coordinates and $\vec{r}$ to the comoving coordinates with the expansion (§2). In our onion model, due to its closed geometry, the scale factor at a given instant equates to the radius of the given hypersphere space slice at that time (fig. 2).

Contrary to ΛCDM, which is built around the idea that matter and energy evolve following Einstein field equations of General Relativity, this works avoids making any premises about the energy contents of the Universe and their interactions. This means that instead of inferring the form of the scale factor from the Friedmann equations (§§2.2) (which are derived by imposing GR over the FLRW metric [2]), we will make a geometrically motivated hypothesis about the form of the scale factor and build the theory around it. This can be understood as follows. ΛCDM is constructed around the evolution of matter and energy (given by GR) and states the

geometry of the Universe (given by the scale factor) as a consequence of this evolution (13). In our model, the approach is opposite. We build a theory around a hypothesis for the geometry of the Universe, and assert that the evolution of its components must conform to it (18).

The precise form of the scale factor can be narrowed down for a closed universe as spacetime must be smooth (i.e. a 4-dimensional manifold). It must admit a tangent space at the BB, which means its geometry must resemble the Euclidean space $\mathbb{R}^4$ at that point, and therefore $R(t) \xrightarrow{t \to 0} ct$. Having homogeneity and isotropy, spacetime must be one of three constant curvature 4-dimensional manifolds ($S^4, \mathbb{R}^4, H^4$) [13], [14], and accordingly the only possibilities for the scale factor are;

$$R(t) = \begin{cases} cT\sin(\frac{t}{T}) & \text{if } k_{st} > 0 \\ ct & \text{if } k_{st} = 0 \\ cT\sinh(\frac{t}{T}) & \text{if } k_{st} < 0 \end{cases} \tag{3}$$

The different expressions for the scale factor correspond to the spatial 3-hypersphere (k=1) expanding on different geometries of 4-dimensional spacetime. This spacetime geometries can be: a 4-dimensional hypersphere (de Sitter space) S^4 ($k_{st} > 0$, fig. 2c), the Minkowski spacetime $\mathbb{R}^4$ ($k_{st} = 0$, fig. 2b), and a 4-dimensional hyperboloid (anti-de Sitter space) H^4 ($k_{st} < 0$, fig. 2a). k_{st} is therefore defined as the constant curvature of the 4-dimensional manifold representing spacetime. Correspondingly, the constant T is related to the 4-D manifold radius of curvature.

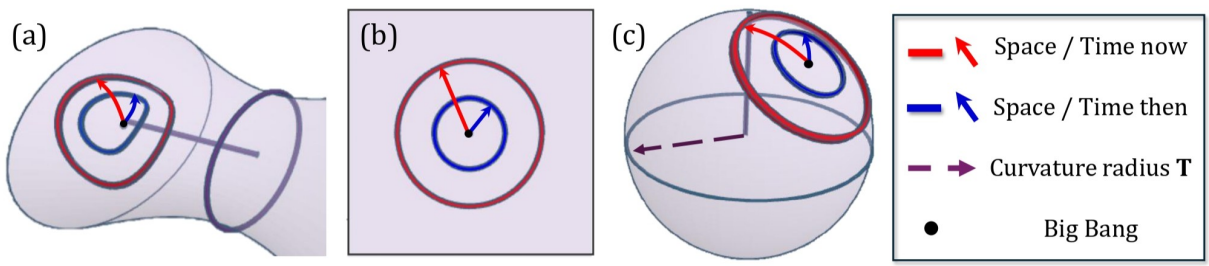


Figure 2: 1+1D diagrams for the three possibilities of spacetime geometries under homogeneity and isotropy. Two different space slices are shown in each panel (red and blue solid lines) for two different cosmological times (red and blue arrows). Left panel (a) corresponds to the hyperboloidal geometry ($k_{st} < 0$). Center panel (b) corresponds to the flat geometry ($k_{st} = 0$). Right panel (c) corresponds to the spherical geometry ($k_{st} > 0$). The curvature radius of spacetime T is shown as a purple dashed line (only in panel (c) for visualization reasons)

Among these parametrizations for the scale factor, the one that has been most extensively studied in the literature is the linear (coasting) model. It has been proved that the linear model provides an elegant solution to several conundrums of the ΛCDM model. This model does not suffer from the *cosmological constant problem*, in the sense that the energy density reaches the correct value at the Planck scale [16], [17], [18] (as opposed to ΛCDM, where the vacuum energy differs in 120 orders of magnitude of the value needed to account for DE [19]). As

noted by [15], the trajectories of the light rays defining the past and future light cones in a closed cosmology describe logarithmic spirals (yellow dotted lines in fig. 1) that turn around the whole universe infinitely many times, giving an explanation to the *horizon problem*. Also, as in these models space forms a closed hypersphere, no fine-tuning is needed to guarantee the geometry of space, and the *flatness problem* disappears.

While a detailed account of the composition (matter, radiation, dark energy) of the Universe in this model is beyond the scope of this work, plausible scenarios for nucleosynthesis [16], the linear growth of structure [20], and the matter power spectrum (compared to CMB anisotropies and the Lyman-α forest) [21] have been formulated (although exclusively in the context of coasting cosmologies). In the present work, we focus on the geometrical constraints imposed by state-of-the-art measurements of the different cosmic distances, for our finite universe with *R(t)* given by (3), the **onion model**.

2 Methods

2.1 Theoretical foundations

FLRW metric

All modern cosmologies are built around the *Cosmological Principle*. This empirical statement establishes that the Universe must be spatially homogeneous and isotropic at large scales. The assumption was first vaguely made by *Einstein* [1] in 1917, and then formalised and named by *Milne* [22](pg24) in 1935. The homogeneity of space indicates that, at every instant of time, all the spacial locations in the Universe are equivalent (e.g. they have exactly the same density), while isotropy refers to the equivalence of all directions. However, this cannot hold for *spacetime*, as the mere existence of a BB poses a privileged point in time.

It can be shown that, starting from this principle without any additional assumptions, one arrives uniquely at the Friedmann-Lemaitre-Robertson-Walker metric (the mathematical description of spacetime's fabric) [23](Ch13.5). The first formulation of this metric was done by *Alexander Friedmann* in 1922 [2] (extended in 1924 to universes of negative curvature [3]) as a solution to Einstein's field equations. *Georges Lemaitre* in 1927 did the first derivation of the metric from solely homogeneity and isotropy [4] and in the 1930's *Howard Robertson* and *Arthur Walker* proved that, up to a coordinate transformation, no other metric satisfies these conditions [13], [14]. The standard expression for this metric has been given in (1).

Cosmological redshift

Given that the scale factor *R(t)* can vary with time, it is expected that the wavelength of a photon an observer measures from a distant emitter is shifted according to the changes in *R(t)*. This effect is called **cosmological redshift (z)**, and is of essential importance in the context

of cosmology. In a FLRW universe it adopts the following simple expression;

$$\frac{1}{1+z} = \frac{R(t)}{R(t_0)} \equiv a(t) \tag{4}$$

where we have defined the quantity *a(t)* as the *adimensional scale factor* ($R(t_0) \equiv R_0$ corresponds to the value of the scale factor at the present day). The derivation of this expression can be found in [24](Ch1.2).

For $t \simeq t_0$, we may expand *a(t)* in a power series; $a(t) \simeq 1 + c(t - t_0)H_0 + ...$ Here, we have then defined H_0, the *Hubble constant*, as $H_0 \equiv \dot{a}(t = t_0)$ and therefore can write the lineal expansion for equation (4), known as *Hubble's law*, as;

$$z \simeq H_0 \cdot d + ... \tag{5}$$

being d the proper distance between two observers ($d = c(t - t_0)$).

To conclude this discussion on *cosmological redshift*, it is also important to consider the concept of *peculiar velocity*. In addition to the motion driven by the expansion of the Universe, astrophysical objects exhibit their own motion due to the gravitational attraction of nearby masses. For objects relatively close to Earth (z<0.1), these peculiar velocities can be comparable to the velocity from cosmic expansion, leading to a *Doppler shift* that mimics or competes with the *cosmological redshift*. This introduces a challenge in accurately measuring this essential observable for nearby objects.

Cosmic distances

From the definition of the metric various cosmological distances can be defined, depending on the particular case under study. We will now discuss those most relevant to this work.

We sill start with the radial comoving distance (D_c), defined between objects static with respect to the expansion ($ds = 0$, $d\theta = 0$, $d\varphi = 0$). Integrating between an emitter at *t* (*z*) and nowadays ($t = t_0, z = 0$);

$$D_c = R_0 \int_t^{t_0} \frac{cdt'}{R(t')} = \int_0^z \frac{cdz'}{H(z')} \tag{6}$$

where he have introduced the **Hubble parameter**, $H(t) \equiv \frac{\dot{R}(t)}{R(t)}$. From here, we can define the *metric distance* (also known as *comoving transverse distance*), $D_M(t)$, as the distance between two objects measured within the same cosmic time ($dt = 0$). As shown in [25], its expression depends on the spatial curvature of the Universe:

$$D_M(t) = \begin{cases} \frac{R_0}{\sqrt{k}} \sin(\frac{D_c(t)\sqrt{k}}{R_0}) & \text{if } k > 0 \\ D_c(t) & \text{if } k = 0 \\ \frac{R_0}{\sqrt{|k|}} \sinh(\frac{D_c(t)\sqrt{|k|}}{R_0}) & \text{if } k < 0 \end{cases} \tag{7}$$

We can also estimate the distance to an object by measuring the flux (total energy per second per unit area, *F*) received. For an emitter with absolute luminosity L, we can write [24](Ch1.3);

$$F = \frac{L}{4\pi D_L^2} \tag{8}$$

where we have introduced the *luminosity distance* D_L. This distance is related to the *comoving traverse distance* by $D_L = D_M(1+z)$, since the rate at which photons arrive to the observer is reduced by a factor *a(t)* compared to the emitter, due to the expansion of the Universe [24](Ch1.4). Likewise, another estimation of distance can be made by measuring the angular separation of an object, θ. The proper diameter of an object at $t = t_1$, $r = r_1$ is $\ell(t_1) = R(t_1)r_1 \sin\theta \xrightarrow{\theta<<1} R(t_1)r_1\theta$. Defining the *angular diameter distance* as;

$$D_A \equiv \frac{\ell}{\theta} \tag{9}$$

and knowing $D_M = R(t_0)r_1$ we find [23](Ch14.4);

$$D_M = D_A(1+z) = \frac{D_L}{1+z} \tag{10}$$

Also relevant to this work are the *Hubble distance* (D_H) and the *spherically-averaged distance* (D_V), defined as follows:

$$D_H = \frac{c}{H(z)} \tag{11}$$

$$D_V = (zD_M^2 D_H)^{1/3} \tag{12}$$

Next, we are going to study how the predictions for these observables are different between ΛCDM and the onion model.

2.2 Model comparison

ΛCDM

In ΛCDM the evolution of the scale factor, and therefore of the Universe, is given by the Friedmann equations, first derived by Friedmann (1922) [2]. These equations are obtained

by substituting the FLRW metric into Einstein's field equations under the assumption that the components of the Universe behave as a perfect fluid of density ρ and pressure p (for a detailed derivation see [23](Ch15.1)).

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{c^2 k}{R^2} + \frac{c^2\Lambda}{3} \tag{13a}$$

$$\dot{H} + H^2 = \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + \frac{3p}{c^2}) + \frac{c^2\Lambda}{3} \tag{13b}$$

It is also common, in the context of ΛCDM, to rewrite this equation by introducing a density parameter for each component of the universe;

$$\Omega_i(t) \equiv \frac{\rho_i(t)}{\rho_c(t)}, \quad \text{where} \quad \rho_c(t) \equiv \frac{3H^2(t)}{8\pi G} \tag{14}$$

When modelling the different components as perfect fluids and taking their respective equations of state ($p(t) = c^2 w\rho$; $\rho \propto a^{-3(1+w)}$) into account [23](Ch15), we arrive, for the first Friedmann equation (13), at;

$$H^2 = H_0^2\left[\Omega_{r,0}a^{-4} + \Omega_{m,0}a^{-3} + \Omega_{k,0}a^{-2} + \Omega_{\Lambda,0}\right] \tag{15}$$

where $\Omega_{r,0}$ and $\Omega_{m,0}$, stand respectively for the density parameters of radiation and matter at present time (t_0); $\Omega_{k,0} \equiv -\frac{c^2 k}{R_0^2 H_0^2}$ can be interpreted as a *density of curvature* at the present day; and $\Omega_{\Lambda,0} \equiv \frac{c^2\Lambda}{3H_0^2}$ as the density associated to dark energy (a mysterious fluid of negative pressure with its origin in the cosmological constant Λ), which is constant throughout the history of the Universe (in ΛCDM). It is also useful to define $E^2(t) = [\Omega_{r,0}a^{-4} + \Omega_{m,0}a^{-3} + \Omega_{k,0}a^{-2} + \Omega_{\Lambda,0}]$

In terms of these density parameters we can write a *closure relation* [23](Ch1.5). For $t = t_0$ the first Friedmann equation (13) is reduced to;

$$1 = \Omega_0 - \frac{c^2 k}{R_0^2 H_0^2} = \Omega_0 + \Omega_{k,0} \tag{16}$$

with $\Omega_0 = \Omega_{r,0} + \Omega_{m,0} + \Omega_{\Lambda,0}$, showing that for a flat universe as ΛCDM, $\Omega_0 = 1$.

With this in mind, one can integrate the distances defined in §§2.1 in terms of the density parameters, arriving to;

$$D_c = \frac{c}{H_0}\int_0^z \frac{dz'}{E(z')} \tag{17}$$

In this work, the computation of the distances has been carried out by numerical integration using simple *Euler integration*. The data used in these calculations comes from the *Planck18* collaboration [9] (implemented in *Astropy v5.2.1*), as a representative benchmark of the ΛCDM model. Only three free parameters are needed in our implementation; the *Hubble constant* $\mathbf{H_0}$,

the dark energy density parameter $\mathbf{\Omega_{\Lambda,0}}$ (referred to present time unless explicitly indicated Ω_Λ(t)), and the *Baryonic Acoustic Oscillations scale radius* $\mathbf{r_d}$ (see §§2.3). In ΛCDM the Universe is flat, so Ω_k can be set to strictly zero, and at present time the contribution of radiation to the energy density of the Universe is negligible so $\Omega_{r,0} \approx 0$. With this in mind, and using the closure relation (16) the matter density parameter can be calculated $\Omega_{m,0} = 1 - \Omega_{\Lambda,0}$.

Onion model

As mentioned in §§1.3, the approach in the *onion model* differs from that of ΛCDM in that its hypothesis is purely geometrical, without any assumptions about the Universe's components. This implies that we cannot rely on the Friedmann equations (13) for the evolution of the Universe as they are formulated based on GR and the assumption that the Universe behaves as a perfect fluid. We have to obtain our own evolution equations directly from the form of the scale factor given in (3). Taking its time derivative and after some algebra one arrives at;

$$H^2(t) = \begin{cases} \dfrac{c^2}{R^2(t)} - \dfrac{1}{T^2} & = H_0^2\left[\dfrac{c^2a^{-2}}{\dot{R}^2(t_0)} - \dfrac{1}{T^2H_0^2}\right] & = H_0^2\left[-\Omega_{k,0}a^{-2} + \Omega_{\Lambda,0}\right] & \text{if } k_{st} > 0 \\ \dfrac{c^2}{R^2(t)} & = H_0^2\left[\dfrac{c^2a^{-2}}{\dot{R}^2(t_0)}\right] & = H_0^2\left[-\Omega_{k,0}a^{-2}\right] & \text{if } k_{st} = 0 \\ \dfrac{c^2}{R^2(t)} + \dfrac{1}{T^2} & = H_0^2\left[\dfrac{c^2a^{-2}}{\dot{R}^2(t_0)} + \dfrac{1}{T^2H_0^2}\right] & = H_0^2\left[-\Omega_{k,0}a^{-2} + \Omega_{\Lambda,0}\right] & \text{if } k_{st} < 0 \end{cases} \quad (18)$$

By analogy with ΛCDM, the *curvature density parameter* can be defined as $\Omega_{k,0} \equiv -\frac{c^2}{\dot{R}^2(t_0)}$, since it appears as the coefficient of the term proportional to a^{-2} in the evolution equation. Taking into account that the onion universe is closed (k=1), this expression is identical to the one defined for ΛCDM in (15). Extending the comparison, we define $\Omega_{\Lambda,0} \equiv \mp\frac{1}{T^2H_0^2}$, for $k_{st} > 0$ and $k_{st} < 0$ respectively (and obviously $\Omega_{\Lambda,0} = 0$ for $k_{st} = 0$). Crucially, in the onion model, Ω_Λ has **no relation** with any *cosmological constant* Λ, but just appears as a direct consequence of the curvature of spacetime, controlled by its radius *T*. This gives a physically-motivated explanation to the constant term in the evolution equation for the scale factor, and does not require the introduction of any *negative pressure, mysterious fluid* to the model. In general, equation (18) shows the evolution of $R(t)$ to be a consequence of the geometry of spacetime, and not of the energy components of the Universe. This formulation leaves us with the following expression for the *closure relation* within the onion model;

$$1 = \Omega_\Lambda(t) - \Omega_k(t) \quad ; \quad \Omega_i(t) \equiv \frac{\rho_i(t)}{\rho_c(t)} = \frac{\Omega_{i,0}}{\Omega_{\Lambda,0} - \Omega_{k,0}a^{-2}} \quad (19)$$

Those readers with a more mathematical inclination might have already realized how the hypothesis for the scale factor (3) calls for a unified definition of the model, by letting the curvature radius *T* take real and purely imaginary values. This is a well known analytical continuation that transforms the space S^4 (positive curvature) into H^4 (negative curva-

ture) [23](Ch13). In our case, it provides a unique definition for $R(t) = cT\sin(t/T)$, with $T \in (0, +\infty) \cup i(0, +\infty)$ (as $\sin(ix) = i\sinh(x)$). This covers the three geometrical cases of interest: positive curvature (real T), flat space (formally $T \to \infty$), and negative curvature (purely imaginary T). It can also be shown how the previous results for the evolution equation and the energy density parameters can also be unified under a unique expression for the three spacetime geometries.

The computation of the cosmological distances can be done analytically in the onion model as we have a simple expression for $R(t)$. Substituting (3) into definition (6) we get;

$$D_c(t) = R_0 \cdot \begin{cases} \ln\left(\dfrac{\tan(t_0/2T)}{\tan(t/2T)}\right) & \text{if } k_{st} > 0 \\ \ln\left(t_0/t\right) & \text{if } k_{st} = 0 \\ \ln\left(\dfrac{\tanh(t_0/2T)}{\tanh(t/2T)}\right) & \text{if } k_{st} < 0 \end{cases} \tag{20}$$

and the other cosmological distances can be obtained directly by use of the definitions (7-12).

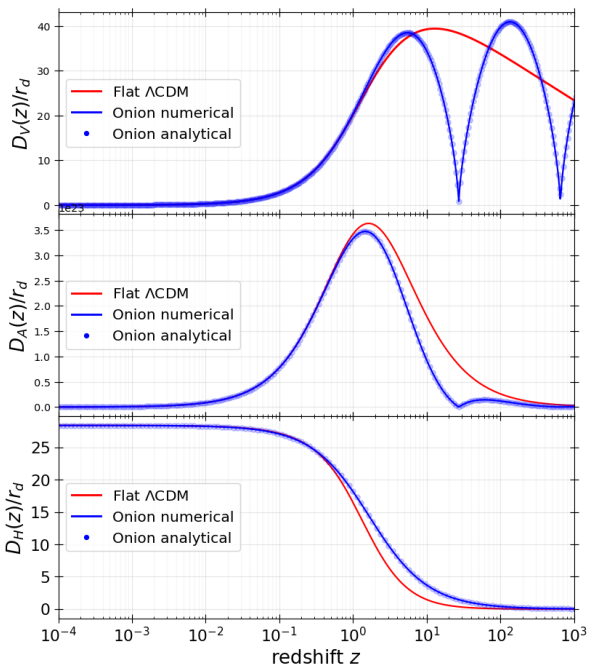


Figure 3: Evolution of cosmological distances with redshift for ΛCDM (red) and onion (blue) models. For onion, a comparison between analytical (blue dots) and numerical (blue solid line) computations is shown. From top to bottom, *spherically-averaged* (D_V), *angular diameter* (D_A), and *Hubble* (D_H) distances; normalized by r_d.

However, within this work the computation has been implemented numerically in order to use the same framework as for ΛCDM. For analogy, the same three free parameters have been used in the computations; $\mathbf{H_0}$, $\mathbf{\Omega_{\Lambda,0}}$, and $\mathbf{r_d}$. The energy density parameter associated to curvature $\Omega_{k,0}$ is fixed by the closure relation (19), and no other density parameters appear in the theory as no predictions on energy components are made. Tests have been run contrasting the numerical integration with the analytical expressions, and a comparison of both of these calculations with the predictions of ΛCDM can be found in fig. 3.

Within the calculations of the onion model (blue solid and dotted lines), no difference can be noted between the numerical and analytical calculations, proving the proper functioning of the numerical integration. In regards to the comparison with ΛCDM, although the differences between both models are significant at large scales, in the redshift

range where data is available at present time ($z < 5$), behaviours are similar. While this means that with available data it is difficult to discriminate between both models, it opens the door for a clear distinction in the future when observations are made for higher redshifts. One of the most evident deviations happens around redshift $z \simeq 28$, where the onion model predicts a zero in the *comoving traverse distance* (which translates to a zero in D_A, D_V). If an object were to be measured at such redshift, according to the onion model, it should span the whole sky. Nevertheless, the probability for this is minimal, as the volume subtended by the corresponding region of spacetime also tends to zero, so it would probably be empty.

2.3 Observables and datasets

In this work, we have limited ourselves to the use of the two most standardized distance estimators in modern cosmology, *Type IA Supernovae* (SNe) and the measure of the *Baryonic Acoustic Oscillations* (BAO).

Standard candles

Type IA Supernovae are cosmological distance estimators that fall within the category of standard candles as they provide a measure of the *luminosity distance* D_L. They were the distance estimators that first allowed the discovery of the accelerated expansion of the Universe in 1998 [7], [8]. Type Ia Supernovae are among the most luminous astronomical events that exist, with a typical peak absolute magnitude comparable to that estimated for our galaxy. These supernovae occur when a white dwarf in a binary system has accreted enough mass to surpass the *Chandrasekhar limit*, triggering an explosion [24](Ch11.3). Notably, the nature of this explosion is largely independent of the time at which it occurs in the universe's history, the initial mass of the white dwarf, or the type of companion star involved. However, the metallicity (proportion of elements heavier than Helium) of the white dwarf, may influence the explosion, with variations based on the epoch in which the event takes place. This variation in the absolute luminosity of the SNe is mostly correctable through empirical methods. It seems to be well correlated with the light curve of the SNe: the slower the decline time of the SNe light, the higher the absolute luminosity [26]. Once an estimation for the luminosity has been obtained, the luminosity distance D_L can be cleared from expression (8) if the energy flux is measured. A more detailed discussion on the use of *Type IA Supernovae* as distance estimators can be found in [27](Ch3).

In this work we have used the two independent sets of *Type IA Supernovae* for the elaboration of a *Hubble-Lemaitre diagram* (H-L diagram; D_L vs z) and comparison with the models. The first one is the *Pantheon+* compilation (PPlus) [28], which consists of 1550 spectroscopically-confirmed SNe in the redshift range $0.001 < z < 2.26$. This data set has been filtered to keep only objects at a redshift $z > 0.01$ in order to neglect the effect of their peculiar velocity as

discussed in §§2.1, [29]. For the computation of the likelihood, the full statistical and systematic covariance from [30] has been used (see §§2.4 for in depth explanation). We have also used the 1635 SNe sample of the *Dark Energy Survey Year 5* (DES) release [31] with redshifts $0.1 < z < 1.3$, which is independent to the *Pantheon+* dataset. This set has been filtered to neglect faulty data points. The estimation of the likelihood has also been computed using the full covariance matrix (systematic + statistical) provided.

Standard rulers

A *standard ruler* refers to an object of known size that can be used as a distance estimator by measuring the angle it subtends in the sky (by the use of expression (9)). The most used standard rulers are the *Baryon Acoustic Oscillations*. Their existence and possible application as a distance estimators was first proposed in 1966 [32] and thoroughly studied from the second half of the 1990's [33],[34] onwards. However, it was not until 2005 that they were convincingly measured by the *SDSS collaboration* [35].

The BAO measurement method for distance estimations exploits a characteristic length scale imprinted in the matter distribution of the Universe, originated from pressure (*acoustic*) oscillations in the photon-baryon plasma prior to recombination. These oscillations, often referred to as *sound waves*, result from the competition between gravitational infall and photon pressure in the early Universe. Overdensities associated to dark matter in the primordial plasma generated spherical pressure waves, which propagated outward at the relativistic sound speed (c_s) until the epoch of recombination ($z = z_*$), when photons decoupled from baryons and the waves ceased to propagate. The maximum comoving distance traversed by these waves before decoupling defines the acoustic scale (r_d), a standard ruler that is preserved in the late-time large-scale structure. By measuring this scale in the distribution of galaxies at different redshifts, one can infer both the angular diameter distance $D_A(z)$ and the Hubble parameter $H(z)$ (or equivalently D_H). For some datasets with a lower resolution (e.g. photometric redshift estimators) it is only possible to get a combined measure of both, $D_V(z)$. The acoustic scale can be computed theoretically as the comoving distance travelled by sound waves from the Big Bang until recombination, and its precise value depends on the physical conditions of the early Universe, such as the baryon density and the matter-radiation content [27](Ch4).

$$r_d = \int_0^{t_*} \frac{c_s(t)}{a(t)} dt = \int_{z_*}^{\infty} \frac{c_s(z)}{H(z)} dz \tag{21}$$

This calculation requires assumptions of the energy-matter content of the early Universe and exceeds the objectives of this work. In our case, r_d will be left as a free parameter of the model.

The BAO datasets that have been used in this work can be divided into three categories attending to the cosmological distance measured:

- $\mathbf{D_V}$: SDSS [36](Tbl3) , WiggleZ [37](Tbl1) , DESI DR1 [38](Tbl1), 6dFGS [39]
- $\mathbf{D_A}$: SDSS [36](Tbl3) , WiggleZ [40](Tbl4) , DESI DR1 [38](Tbl1), DES YR3 [41](Tbl2), DES YR6 [42]
- $\mathbf{D_H}$: SDSS [36](Tbl3) , WiggleZ [40](Tbl4) , DESI DR1 [38](Tbl1)

Three important considerations must be highlighted regarding the treatment of BAO in relation with this work. Firstly, it is important to mention how if r_d is treated as a free parameter, estimations of H_0 become degenerate with its value. This arises from the fact that line-of-sight separations depend on $H(z)r_d$, while transverse separations depend on $D_A(z)/r_d$ [27](Ch4.1). Secondly, since the BAO signal is intrinsically weak on large scales, mapping vast volumes of the Universe is required to detect it and obtain precise distance measurements. This represents a challenge at high redshifts, where wide redshift bins are necessary to include a statistically significant number of galaxies capable of revealing the signal [27](Ch4.4), as the flux measured from distant objects is limited. As discussed in §4, this issue has been noted and taken into account in the parameter estimations presented in this work. Finally, it is important to mention potential systematic biases associated with the fiducial cosmology assumed in certain computations, which may affect the analysis of galaxy distribution functions [27](Ch4.5,Ch7.3).

2.4 Statistical analysis

The ultimate goal of this work is to perform a statistical test between each models predictions for cosmological distances (§§2.2) and the corresponding observational data (§§2.3). The cosmological parameters defining each model (H_0, $\Omega_{\Lambda,0}$, r_d) have been adjusted through a grid search to identify their best-fit values. Subsequently, these optimal parameter values have been used to compare both models, assessing their compatibility with the observational data. In this context, a formal and meticulous statistical analysis has proven to be fundamental.

Chi-squared test χ^2

The cosmological parameter estimations carried out in this work have been performed using a standard chi-squared (χ^2) minimization procedure. This approach, thoroughly discussed in [43](Ch6), is widely adopted in cosmological data analysis due to its effectiveness in handling Gaussian-distributed observational uncertainties. The quantity minimized is given by:

$$\chi^2 = \sum_{i=1}^{N} \frac{[y_i - f(x_i)]^2}{\sigma_{y,i}^2} \tag{22}$$

where y_i are the observed data points, x_i the corresponding independent variables, $\sigma_{y,i}$ the associated uncertainties, $f(x_i)$ the model prediction evaluated at those points, and N the size

of the dataset. This quantity indicates how well the model's predictions fit the observed data. In this work, it is important to note that we use an averaged χ^2, where the result given by (22) is divided by the total number of points N, or equivalently, we calculate the mean χ^2.

Equation (22) can also be extended to datasets that provide a covariance matrix $\hat{C} = \hat{C}_{sys} + \hat{C}_{stat}$, a $N \times N$ matrix which accounts for the statistical and systematic errors between data points;

$$\chi^2 = (\mathbf{y} - f(\mathbf{x}))^T \hat{C}^{-1} (\mathbf{y} - f(\mathbf{x})) = \sum_{i,j} [y_i - f(x_i)] C_{ij}^{-1} [y_j - f(x_j)] \tag{23}$$

In this notation $\hat{C}^{-1}$ is the inverse of the covariance matrix, and its diagonal elements C_{ii}^{-1} are the uncertainties of each data point $\sigma_{y,i}^2$, while off-diagonal elements represent the correlation between measurements i and j. We have implemented this analysis to the SNe datasets.

Nevertheless, we found that this approach was not sufficient, and it was crucial to implement the treatment of the uncertainty associated with the dependent variable for some of our observables (it failed mainly for high-redshift BAO measurements). This means extending (22) to include uncertainties in the dependent variable $\sigma_{x,i}$. We based our approach on Orthogonal Distance Regression (ODR), of which Total Least Squares (TLS) is a special case for linear models [43](Ch6). We extended this treatment to non-linear models [43](Ch8). ODR allows for the simultaneous minimization of both x and y residuals, adjusting for uncertainties in both variables by minimizing the orthogonal distances from data points to the model's curve by incorporating the model's variation $\frac{\partial f}{\partial x_i}$. In this case, we can compute χ^2 as;

$$\chi^2 = \sum_{i=1}^{N} \frac{[y_i - f(x_i)]^2}{\sigma_{y,i}^2 + \sigma_{x,i}^2 \left(\frac{\partial f}{\partial x}\right|_{x=x_i}^2} \tag{24}$$

This can be trivially extended to include the covariance matrix (23) as it just affects its diagonal elements.

In this work, we have taken $\sigma_{x,i}$ to be one over the squared root of three of the total error given for each x_i. This comes from the standard result in the context of Gaussian-distributed errors, where $\sigma_{x,i} \approx \frac{\Delta x_i}{\sqrt{3}}$, being Δx_i the total uncertainty.

For m independent datasets ($i = 1, 2, ..., m$), the combined χ_T^2 can be obtained as the sum of the χ_i^2 of each dataset. If we define the *likelihood* of each model as $\mathcal{L}_i \propto e^{-\chi_i^2/2}$, we can calculate the combined *likelihood* as $\mathcal{L}_T \propto \prod_{j=1}^{m} \mathcal{L}_j$. Being that the $\mathcal{L}_i$ are normalized, after some algebra we can show that [43](Ch10);

$$\chi_T^2 = \sum_{j=1}^{m} \chi_j^2 \tag{25}$$

Bayesian *vs* Frequentist approaches

Besides the statistical considerations in the calculation of χ^2, we also want to address how the parameters have been estimated from these results. For each model, we performed a grid search to find the best fit. However, an important question arises regarding how the *best fit* is defined. We have explored two approaches: *Bayesian* and *Frequentist*.

The *Frequentist* approach defines the best fit as the parameter combination that minimizes the χ^2 value. While this criterion is conceptually the most straightforward, it presents certain limitations. In particular, we observe a significant sensitivity to small variations in the input parameters. This instability stems from the discrete nature of the grid search and the relatively flat behaviour of χ^2 near its minimum, which can lead to abrupt, unwanted shifts due to minor numerical fluctuations or observational uncertainties [44](Ch5). As a result, this approach lacks robustness and may not be not representative of the statistical behaviour of the model.

On the other hand, a *Bayesian* approach defines the best fit as the point in the parameter space with the highest posterior probability, taking into account the likelihood as a normalized weight that we compute as $w(\vec{\theta}) \propto \exp\left[-(\chi^2(\vec{\theta}) - \chi^2_{min})/2\right]$, being $\vec{\theta} = (H_0, \Omega_{\Lambda,0}, r_d)$, our parameters. In this context, the expected value and its uncertainty is given by [44](Ch9);

$$\vec{\theta}_{best} = \sum_{\vec{\theta}} w(\vec{\theta})\vec{\theta} \qquad ; \qquad \sigma_\theta = \sqrt{\sum_{\vec{\theta}} w(\vec{\theta}) \left(\vec{\theta} - \vec{\theta}_{best}\right)^2} \tag{26}$$

This formulation provides a robust alternative to the *Frequentist* method, as it naturally incorporates parameter uncertainties, avoiding abrupt instabilities in the best-fit values.

Throughout this work, primarily a *Bayesian* approach has been used, leading to the results presented below (§3). A comparison with the *Frequentist* approach has also been conducted, resulting in the conclusions that have just been outlined.

3 Results

3.1 Parameter estimations

As mentioned previously, this work can be naturally divided into two distinct phases. The first part consists of the estimation of the cosmological parameters through a direct grid search over the model's parameter space. A grid search is a brute force optimization method in which the relevant parameters are sampled over a regular multidimensional grid. In each point of the grid, an objective function, in this case, a chi-squared estimator is evaluated. The parameter space explored in this work for the three parameters of our models mentioned in §§2.2 (H_0, $\Omega_{\Lambda,0}$, r_d) encompasses the range;

$$55 \leq H_0 \leq 95, \quad -0.95 \leq \Omega_{\Lambda,0} \leq 0.95, \quad 50 \leq r_d \leq 200.$$

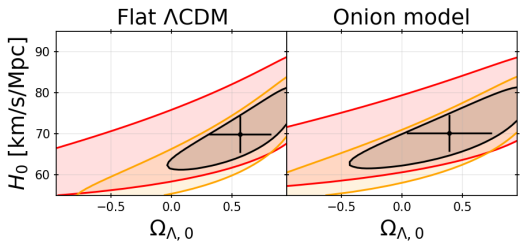


Figure 4: 2D contours (at a 1σ confidence interval from the *Bayesian* minimum) in H_0-$\Omega_{\Lambda,0}$ parameter space for ΛCDM (left) and onion (right). Estimations from *Pantheon+* (red) and *DES* (yellow) SNe measurements. Combined contour and best fit with uncertainties in black.

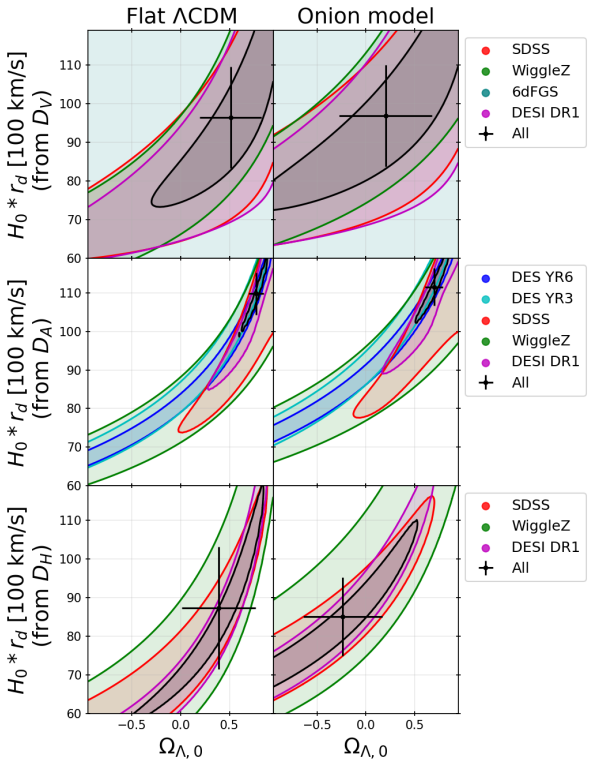


Figure 5: 2D contours (1σ) in $H_0 r_d$-$\Omega_{\Lambda,0}$ parameter space for ΛCDM (left) and onion (right). Only the combined estimator ($H_0 r_d$) can be obtained in 2D as the parameters are degenerate. From top to bottom, panels for estimations from the different observables; *spherically-averaged* (D_V), *angular diameter* (D_A), and *Hubble* (D_H) distance measures. Different colours for every dataset, combined contours and best fits with uncertainties in black.

2D contours

This method has allowed us to compute confidence contours and identify the regions of parameter space that best reproduce the observational data. We have chosen to represent the regions of the parameter space that differ less than 1σ from the expected value, as defined in a *Bayesian* approach (26). To ensure consistency between datasets, two steps have been followed in the estimation of the parameters.

Firstly, a search has been performed independently for each observable. This has allowed us to test the agreement between all the datasets used, and establish a direct comparison between those providing the same measurement. The observables studied in this work (§§2.3) can be organized into two categories: non-degenerate and degenerate estimators. This classification refers to whether the observable allows estimation of only H_0 and $\Omega_{\Lambda,0}$ (such as the D_L measurements from standard candles, which are not influenced by the value of r_d) or whether the estimation of H_0 depends on the value of r_d (as is the case for distance estimates from BAO measurements). In the latter case, it has been necessary to perform a combined $H_0 r_d$ estimation in 2D parameter space.

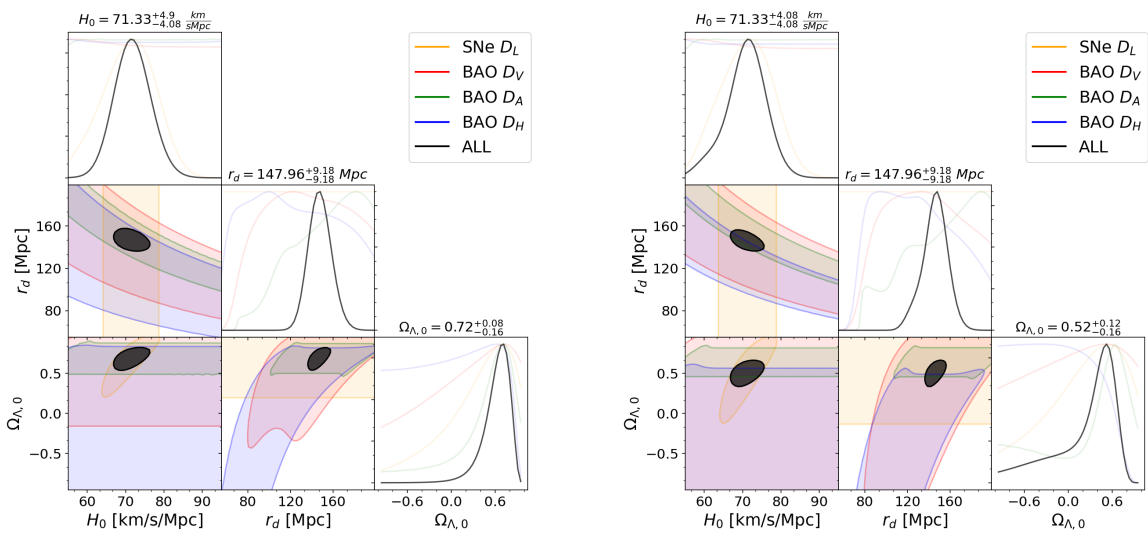
The contours for each dataset are displayed in distinct colours and grouped into separate figures corresponding to each observable. Non-degenerate estimations are presented in the H_0–$\Omega_{\Lambda,0}$ parameter space (fig. 4), while degenerate estimations are expressed in $H_0 r_d$–$\Omega_{\Lambda,0}$ space (fig. 5). Each figure also presents a combined contour (in black) integrating all datasets associated

with each observable using (25), the most probable value and its dispersion (defined in the *Bayesian* sense) is also shown. The estimation of each parameter has been done for both the onion model and ΛCDM, with the results displayed in adjacent columns.

3D contours

Due to the degeneracy inherent to BAO measurements, a secondary analysis has been essential for obtaining independent estimations of the cosmological parameters. This analysis has consisted of projecting the full 3D parameter space onto 2D slices by marginalizing over one parameter at a time. The projection has been carried out within a *Bayesian* framework, assigning each point on the 2D slices the value of the *Bayesian* mean of χ^2 along the third parameter direction (26).

This analysis has been applied to the combined χ^2 values computed for each observable from the previous plots (fig. 4,5). The results are presented in the following corner plots, where every observable is shown in a different colour, and the final combined result is displayed in black. These plots include individual panels for each 2D slice of the parameter space (obtained by marginalizing over one parameter), as well as 1D distributions that provide independent estimations for each parameter (obtained by marginalizing over the other two). Two corner plots are presented in fig. 6, for ΛCDM (left) and the onion model (right).



(a) ΛCDM parameter estimations

(b) Onion model parameter estimations

Figure 6: Corner plots including contours (1σ) of the 3D parameter space (H_0, r_d, $\Omega_{\Lambda,0}$) marginalized over 2D slices for ΛCDM (left) and onion (right) are presented. For every parameter, *Bayesian* 1D probability distribution (with its final value and dispersion) is also presented. Different colours are used for every observable (D_L, D_V, D_A, D_H,) and the combined contour, in black.

The final estimations for the cosmological parameter obtained through this framework are presented in the following table.

Model	$\mathbf{H_0}$ $\left[\frac{\mathrm{km}}{\mathrm{s\,Mpc}}\right]$	$\mathbf{r_d}$ [Mpc]	$\mathbf{\Omega_{\Lambda,0}}$
ΛCDM	$71.3^{+4.9}_{-4.1}$	$148.0^{+9.2}_{-9.2}$	$0.72^{+0.08}_{-0.16}$
Onion	$71.3^{+4.1}_{-4.1}$	$148.0^{+9.2}_{-9.2}$	$0.52^{+0.12}_{-0.16}$

Table 1: Best fit values for the cosmological parameters (H_0, r_d, $\Omega_{\Lambda,0}$) for both models. Errors given by the 1σ dispersion of the fit.

3.2 Model comparison

With the best fit values of the parameters for each model now determined, the second stage of the analysis focuses on evaluating the quality of their predictions. This is done by comparing the minimum χ^2 values obtained for each model across the different observables, thereby assessing their ability to reproduce the data.

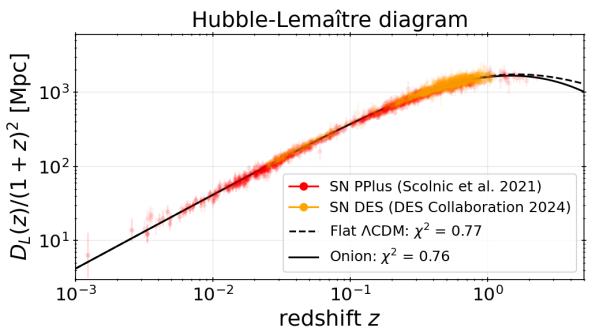


Figure 7: HL diagram (D_L against redshift) comparing ΛCDM (dashed) and onion (solid) with SNe measurements (*Pantheon+* red, *DES* yellow). Parameters (H_0, r_d, $\Omega_{\Lambda,0}$) given by Table 1. χ^2 fit shown in the legend.

For this reason, we have gone back to a separate analysis for each of our four observables. We present separately the widely known *Hubble-Lemaitre* diagram (fig. 7), showing the evolution of D_L with redshift, and the distances estimates using BAO measurements $D_V(z)$, $D_A(z)$, $D_H(z)$ (fig. 8). The predictions of each model (represented by solid and dashed black lines, for ΛCDM and onion models respectively) are shown, alongside the data points from each dataset in different colours and including their error bars.

The χ^2 fits of both models for each observable are presented in the corresponding legend of each panel. By combining the χ^2 values from all observables, we obtain a global assessment of the goodness of fit of both models.

$$\chi^2_{\Lambda CDM} = 1.80 \quad ; \quad \chi^2_{onion} = 3.22$$

4 Discussion

4.1 Parameter estimations

The evaluation of the cosmological parameters has produced results that deserve to be examined in detail. In the 2D analysis of the parameter space, SNe data alone provide weak constraints, especially due to their limited sensitivity to certain cosmological parameters such as $\Omega_{\Lambda,0}$ (fig. 4). Nevertheless, it is worth noting how the SNe datasets used are consistent with each other and provide a combined contour that can serve as a first approximation to the values of the parameters, both in ΛCDM and in the onion models. While some BAO datasets are somewhat more informative in the $H_0 r_d$-$\Omega_{\Lambda,0}$ space, they also yield relatively broad allowed regions when considered independently, with a large degeneration in $\Omega_{\Lambda,0}$ (fig. 5). This is particularly noticeable for older BAO measurements, which contribute little to the overall constraining power due to their larger uncertainties and less precise calibrations. This is evident, for example, in data sets such as *6dFGS* and *WiggleZ*, studies from the early 2010's whose precision is smaller in comparison with recent ones such as *DESI DR1* (2025) and extended versions of *SDSS* (2021). These newer studies provide tighter constraints due to improved precision, and consequently contribute more significantly to the constraining of cosmological parameters. As will be discussed later, a large part of the improved constraining power of these newer datasets is associated with high-redshift measurements. All in all, it is noteworthy that, despite differences in precision and methodology among these datasets, there is strong consistency across them. We observe how the different contours in fig. 5 overlap, forming the combined contour in black. This is a crucial part of the analysis for both the ΛCDM and the onion model, as it demonstrates that the predictions of both models are consistent and in harmony with every dataset. This agreement among tracers enhances confidence in the robustness and reliability of the inferred cosmological parameters, and allows us to proceed with the estimations in 3D

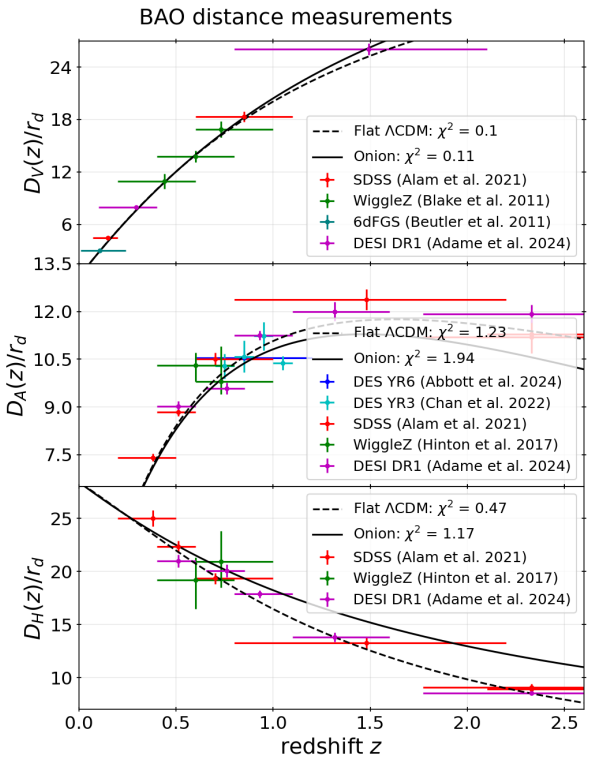


Figure 8: BAO distance measures against redshift comparing ΛCDM (dashed) and onion (solid) with BAO datasets (in different colours). From top to bottom, D_V, D_A, and D_H; normalized by r_d. Parameters (H_0, r_d, $\Omega_{\Lambda,0}$) given by Table 1. The χ^2 fit shown in the legend.

parameter space, where only the combined contours for each observable shall be studied.

The analysis of the various projections of the full contours from the 3D parameter space allows us to observe the consistency among different cosmological observables. From the corner plots several observations can be raised (fig. 6). Although already discussed, we wish to start our analysis noting that supernova data does not constrain the sound horizon scale r_d, since the measurement of the luminosity distance D_L is independent of this parameter (as illustrated in fig. 4). Consequently, the contours associated with this observable in the projections involving r_d are extended across its entire parameter range. This is not the case for BAO measurements, from which we can independently distinguish the parameters H_0 and r_d through these projections. Nevertheless, we confirm the point raised for the 2D contours: BAO distance estimates provide a really loose restriction on the cosmological parameters. The contours shown in fig. 6 only represent 1σ deviations from the minimum χ^2 value, yet, for example for $\Omega_{\Lambda,0}$, they still span almost the entire parameter range in both ΛCDM and onion, especially for D_H measurements. For the two other degenerate parameters (H_0, r_d), this limited constraining power is also reflected. While BAO measurements contribute significantly to the analysis by enabling an estimation of r_d, they fall short of delivering independent and robust constraints on the full set of cosmological parameters. It is only through the combination of SNe and BAO data (represented by the black contour in fig. 6) that we obtain meaningful constraints. This joint analysis effectively breaks parameter degeneracies and yields estimates that are narrower and therefore more informative, making them valuable for cosmological interpretation.

The final results for the parameter estimates also deserve their own analysis (table 1). For ΛCDM we have been able to recover the well established values for H_0, r_d and $\Omega_{\Lambda,0}$ consistent with the literature [9]. This unambiguously validates the effectiveness of our employed numerical framework and statistical analysis. For the onion model, we want to the address the similarity of the values for H_0 and r_d to those of ΛCDM. Although such an outcome was not necessarily expected given the different nature of the models, it is reassuring to find that the onion model does not imply significant deviations in the values of the parameters. This is relevant, as substantial shifts could lead to broader phenomenological implications or potential tensions with other observational constraints. Regarding the value of the density parameter $\Omega_{\Lambda,0}$, we hypothesised a positive value, close to that found in ΛCDM. This expectation stems from the fact that, despite the fundamentally different interpretations of $\Omega_{\Lambda,0}$ in the two models, it remains the term responsible for the linear evolution of the Universe (15, 18). Observing table 1, we see that this result has been confirmed. Phenomenologically, this corresponds to a universe with a negative spacetime curvature (fig. 2a) driven by a hyperbolic sine expansion (3). This evolution of the scale factor resembles the accelerated expansion predicted by ΛCDM, showing the viability of explaining cosmic acceleration through purely geometric arguments without invoking dark energy.

That said, these results should be approached with due caution. When considering the

uncertainties associated with each parameter estimate, the primary conclusion to be drawn concerns the limitations of this purely geometrical approach. The 1σ confidence intervals presented span roughly 5–10% of the corresponding parameter values, reaching as high as 30% for the $\Omega_{\Lambda,0}$ estimation in the onion framework. This high dispersion indicates how the geometrical analysis alone is insufficient to tightly constrain the cosmological parameters. Still, it remains valuable for qualitative and phenomenological studies of the models. It is also worth noting how a significant portion of the observed dispersion stems from certain observables, most notably D_H and D_A, whose individual constraints are rather weak, a limitation that carries over into the combined estimations (fig. 6).

Additionally, it is worth addressing a mild tension, below the 1σ level, in the estimation of $\Omega_{\Lambda,0}$ from D_A and D_H measurements within the onion model. As shown in fig. 6, the corresponding 1σ contours only exhibit a narrow region of overlap in parameter space, which could be interpreted as a potential point of contention for the model. However, the fact that the discrepancy lies within the 1σ confidence region, is enough of an statistical argument to disregard it. Even so, a dedicated analysis was conducted to research the potential origin of this deviation. We found it to be directly correlated with high-redshift points in D_H measurements from *SDSS* and *DESI DR1*. These specific points in these surveys are related to quasar objects (as opposed to the majority of the data, which comes from galaxy samplings) and span broad redshift bins, $\Delta z \sim [1.5,\ 4]$ (§§2.3), while still reporting low dispersions in D_H ($< 5\%$). This heavily influences the χ^2 minimization, ultimately biasing the estimation of $\Omega_{\Lambda,0}$ towards lower values for this observable in the onion model. When these points are excluded from the analysis, the inferred value shifts accordingly and this alleged tension disappears, highlighting the sensitivity of the results to these high-redshift measurements. We expect future observational results at high redshifts help resolve this uncertainty.

Nonetheless, the validity of the cosmological parameter estimates is assessed in the following comparison of the best-fit values obtained for each model.

4.2 Model comparison

The parameter values presented in table 1 yield equally good fits for both the ΛCDM and onion models across the various observables (fig. 7,8). In particular, both models exhibit full consistency with the SNe data (fig. 7), which are widely recognized as the most reliable distance indicators in cosmology. Accordingly, no concerns arise regarding the compatibility of either model with this observable. Similar conclusions hold for BAO datasets. The minimum χ^2 values for D_V measurements are identical for both models. In the comparison of D_A and D_H predictions, one might argue that ΛCDM offers a marginally better fit. Nonetheless, the differences in χ^2 values, on the order of $\sim 1\sigma$, are **not** statistically significant enough to favour one model over the other. Furthermore, this apparent *preference* for ΛCDM vanishes when the considerations raised in the previous section regarding high-redshift measurements are taken

into account. It is also worth noting that these results may also be influenced by ΛCDM being adopted as the fiducial cosmology for the estimations within the BAO surveys (§§2.3).

We have therefore found the geometrical framework provided in this work to be insufficient for discriminating between ΛCDM and onion models using state of the art distance indicators. This conclusion is further supported by the best fit χ^2 values for both models (§§3.2), whose difference is not statistically significant enough to disregard any of the models.

5 Conclusions

The onion universe is a geometrically motivated model which emerges as an alternative to the widely spread ΛCDM, providing different approaches to some of the challenges that arise within this model (§§1.2). Based on the assumption of a closed spatial geometry (k=1) and a scale factor derived from constant spacetime curvature (3), this framework proposes the Universe's evolution dictated solely by geometrical principles, without requiring assumptions about its energy contents (18).

Analytical expressions for the evolution equations of this model (18) and predictions for its main cosmological distances (20) have been derived. A numerical comparison against ΛCDM with observational data from Type Ia Supernovae and Baryon Acoustic Oscillations has been carried out for standard observables (§§2.3). Through a rigorous statistical analysis, inside a *Bayesian* approach and using χ^2 minimization and grid-based parameter searches, estimates for H_0, r_d, and $\Omega_{\Lambda,0}$ have been obtained (§§3.1). The results indicate that no preference can be raised between the onion model and ΛCDM using this purely geometrical approach and up to date datasets (§§3.2).

Remarkably, the model accounts for the accelerated expansion of the Universe without invoking dark energy as a separate physical component. Instead, this phenomenon appears naturally as a consequence of the hyperbolic geometry of spacetime itself, which has been shown to be the best fit to observational data (§§3.1).

However, the considerable uncertainties in the estimation of certain parameters, notably $\Omega_{\Lambda,0}$ within the onion model, underscore the limitations inherent to this study (§§3.1). A natural progression from this work entails abandoning the purely geometrical approach and studying matter and energy evolutions, in order to understand dynamical constraints from early-universe physics, such as nucleosynthesis, CMB anisotropies and structure formation. New high-redshift observations could also help discriminate between ΛCDM and the onion model.

The onion universe represents a viable and conceptually compelling cosmological model. Its consistency with current data, despite only making assumptions on the geometry of spacetime. This motivates a future continuation of this work in order to reduce the uncertainties in its predictions and allow a discrimination between the onion model and ΛCDM.

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